Test report 2 – UI phase A

Browser UI of the parametric CAD (TypeScript/React, CodeMirror 6, three.js) · 2026-10-23 · Commit 15e78ec

End-to-end (Playwright)	Unit tests (Vitest)	First scene	Keystroke → scene
22 / 22 passed	**87 / 87 passed**	**867 ms**	**157 ms**

1 Scope

This report covers the browser UI after phase A: the editor on the left, the 3D view with breadcrumb bar on the right, the status bar at the bottom. The engine (Rust, Wasm, SMLib kernel) runs in a web worker; the UI only interprets the scene description and loads meshes per node on demand.

- **Editor:** Lezer grammar (OpenSCAD with `grid`, `anchor`, `k:`), diagnostics (red underline, line red or yellow, message at the end of the line), inline hints (values, "4 edges"), * lines grayed out, positions of old scenes via `ChangeSet.mapPos`.
- **3D view:** meshes per node with a cache keyed by `mesh_key`, placement via `display_world`, edges as lines, rotate/pan/zoom, "Show all", standard views, axis triad, ground.
- **Code → view:** cursor (node full, rest dimmed), hover over a line, difference/intersection/fillet/chamfer/shell, modifiers `# % ! *`, transformations with coordinate triads before/after and an outline at the original position, 2D shapes as outline with a light fill.
- **View → code:** hovering over a face highlights the line, a click sets the cursor, a double-click on a module instance jumps to the definition.
- **Module context:** `SetContext` when the cursor is in a module definition, breadcrumb bar with a jump one level up.
- **Grids and anchors (read-only):** planes at the edge of the bounding box, intersection points, literal solid / expression dashed, toggle "all grids", hover over a grid expression, anchors as diamonds with an arrow.
- **Files and errors:** autosave to IndexedDB, open/save, examples menu, export STL/3MF, syntax errors and watchdog.

2 Test environment

Operating system	Linux 6.18.44-fc-v140 (x64)
CPU	Intel(R) Xeon(R) Processor @ 2.80GHz · 4 cores
Node.js	v22.22.0
Browser	Chromium 141.0.7390.37 (headless, Playwright 1.56.1)
Graphics	WebGL via SwiftShader (software rendering, no GPU)
Window	1440 × 900 px, device pixel ratio 1
Build	Vite 8.3.5 production build, preview server (vite preview)
Libraries	React 19.3.1, three.js 0.186.1, CodeMirror 6.43.15, Zustand 5.0.16, Vitest 5.0.5

Engine	cad_wasm_bg.wasm 1.16 MiB, smlib.wasm 4.85 MiB (each once in the build)
Source	Branch worktree-agent-a5a96375179179a1c, commit 15e78ec

The end-to-end tests run against the production build. They wait for scene versions and the display state via the test hook `window.__cad`, not for fixed times. Hover and click in the view are real mouse events on the canvas; the hook only supplies a pixel under which a face of the requested line lies.

3 Scenarios (Playwright)

ID	Scenario	Result	Duration	Check
S01	Load: editor, view and first scene	passed	1.9 s	Initial state: spec example gehaeuse.scad in the editor on the left, on the right the scene with ground, axis triad and the anchors of the named instance k (diamonds, with an arrow where dir is set).
S02	Open example	passed	2.3 s	Examples menu → spec/lochplatte.scad: the editor loads the file, after evaluation the view shows the plate with holes, tab and the anchor p.erstes_loch.
S03	Cursor in line: node full, rest dimmed, transformation with coordinate triads	passed	2.2 s	Cursor on translate(k.ecke) in line 22: subtree full at the target, as outline at the original position; coordinate triad dashed before, solid after, arrow from the old to the new origin; rest of the model dimmed.
S04	Pixel check: dimming changes the rendering	passed	2.0 s	Mean brightness of the view without selection 226.3, with cursor on line 22 234.3 (the dimmed rest is brighter).
S05	Module context, difference: subtracted body red	passed	1.9 s	Cursor in line 10 (difference) inside module gehaeuse: context switches to the module (breadcrumb "File › gehaeuse" (German UI: „Datei › gehaeuse“), own coordinate system, grid g shown); result full, subtracted box from line 12 red and semi-transparent.
S06	Hover over line: node semi-transparent	passed	2.0 s	Mouse over line 12 (inner box, subtracted): the body appears semi-transparent in the overall model, visible through the result; cursor is on line 11 (outer box full).
S07	Hover over face highlights line, click sets cursor, double-click jumps to definition	passed	2.8 s	Mouse over the face of the lid at k.ecke: line 22 in the editor is highlighted (face → tri_tags → tag_nodes → node → line). A click sets the cursor in line 22, a double-click on the enclosure jumps into module gehaeuse.
S08	Breadcrumb bar: click jumps one level up	passed	3.5 s	Module context gehaeuse: only the module in its own coordinate system with grid g (intersection points, planes used by the line). A click on "File" sets the cursor on the call k: gehaeuse(w = 60) (line 21) and shows the whole file again.
S09	Modifiers #, %, *, !	passed	3.2 s	24_modifiers.scad: # permanently red and semi-transparent (also as a subtracted body), % gray and semi-transparent as helper geometry, * invisible and line 7 grayed out. With 25_root_modifier.scad the status bar reports the active ! subtree.
S10	Syntax error: last valid state stays visible	passed	2.9 s	Input "]" at the end of line 22: position underlined red, message at the end of the line, status bar "1 error" (German UI: „1 Fehler“); the view keeps showing the last valid state. After deleting it, the model is re-evaluated.

S11	Failed operation: line red, message at end of line	passed	3.3 s	difference(fillet = 20) on a 10 mm cube: fillet failed, line 1 red with a message at the end of the line; the view shows the body without fillet.
S12	Export STL and 3MF	passed	2.5 s	Export STL: gehaeuse.stl, 54 triangles, 2784 bytes (binary, 84 + 50·n). Export 3MF: gehaeuse.3mf (ZIP, 4486 bytes). The status bar reports the export.
S13	Grid: used planes, all grids, hover over grid expression	passed	4.4 s	Module lochplatte, toggle "all grids": planes as thin lines at the edge of the part's bounding box (literal solid, expression dashed), intersection points as dots. Without the toggle only the 6 planes of line 12; hovering over g[loch[i], mitte, boden] highlights the point.
S14	Anchors of named instances	passed	3.2 s	winkel_anker.scad: anchors of the named instances a and b as diamonds, with an arrow where dir is set. Hovering over a.boden_loch highlights this anchor; the cursor on a: winkel() highlights the three anchors of the instance.
S15	Fillet zones blue, selected edges colored, inline hint	passed	3.3 s	fillet_zonen.scad, cursor on fillet(2) in line 12: zones (grid line and sphere) blue and semi-transparent, selected edges orange; inline hint "2 zones · 2 edges" at the end of the line.
S16	Positions of old scenes are carried along while typing	passed	2.9 s	Two lines inserted at the top while the engine is blocked for 1.5 s: the cursor in line 24 still selects the node translate(k.ecke) (formerly line 22) in the old scene – source positions are carried along via ChangeSet.mapPos until the new scene arrives.
S17	Watchdog reports a hung engine	passed	3.1 s	Engine worker blocked for 5 s (test helper, watchdog 0.8 s): the worker is terminated and restarted, the status bar reports the error in red; the last scene stays visible, the next change is evaluated normally.
S18	Autosave to IndexedDB	passed	4.3 s	A change is saved to IndexedDB after 0.4 s; after reloading the page the changed text is back in the editor.
S19	Dark theme and standard views	passed	2.4 s	Dark theme, iso view, cursor on k: gehaeuse(w = 60): instance full, anchors of the instance highlighted, rest dimmed.
S20	intersection and 2D shapes	passed	4.0 s	05_nested_booleans.scad, cursor on intersection(): cube and sphere semi-transparent, result full. 27_spacers_2d.scad, cursor on square([30, 10]): 2D shape as outline with a light fill in its plane (top view).
S21	Open and save without File System Access API (upload and download)	passed	1.3 s	Without the File System Access API: "Open..." (German UI: „Öffnen…“) uploads a file via the file dialog (teil.scad is evaluated), "Save" (German UI: „Speichern“) downloads the source unchanged. With the API (Chromium) the same functions run via showOpenFilePicker/showSaveFilePicker.
S22	Measurements: keystroke to new scene, fps while rotating	passed	12.2 s	Five changes of the literal in line 21: keystroke → new scene drawn, median 157 ms (in-app measurement). Rotation over 120 frames: 14 fps (SwiftShader, no GPU).

4 Unit tests (Vitest)

File	passed	Failed
web/src/editor/decorations.test.ts	6 / 6	–

web/src/editor/lang.test.ts	42 / 42	–
web/src/logic/context.test.ts	5 / 5	–
web/src/logic/display.test.ts	15 / 15	–
web/src/logic/grids.test.ts	5 / 5	–
web/src/logic/positions.test.ts	5 / 5	–
web/src/logic/sceneIndex.test.ts	9 / 9	–

Logic without a browser: Lezer grammar on all examples, mapping code ↔ node (cursor, face → line, ambiguity, module context), display states, grid display, breadcrumb bar and jump to definition, editor decorations, mapping of old positions. Scenes as fixtures from the real engine (`web/scripts/dump-scene.mjs`).

5 Measurements

Quantity	Value	Measurement
Load time until first scene	867 ms	Navigation until the first draw of the complete scene of gehaeuse.scad (loading Wasm, evaluation, tessellation, rendering)
First evaluation	113 ms	SetSource until complete scene (cold cache)
Keystroke → new scene (median)	157 ms	In-app measurement: last change until the new scene is drawn; individual values 73, 131, 127, 250, 165, 157 ms
Keystroke → new scene, test view (median)	315 ms	Playwright: keystroke until the scene is visible in the test hook (incl. 50 ms polling interval)
Evaluation (median)	42 ms	SetSource until complete scene during the changes (warm cache)
Frame rate while rotating	14 fps	120 frames of camera rotation around the z axis, one full render per frame (SwiftShader)
Largest image file in the report	140 KiB	JPEG, quality 82

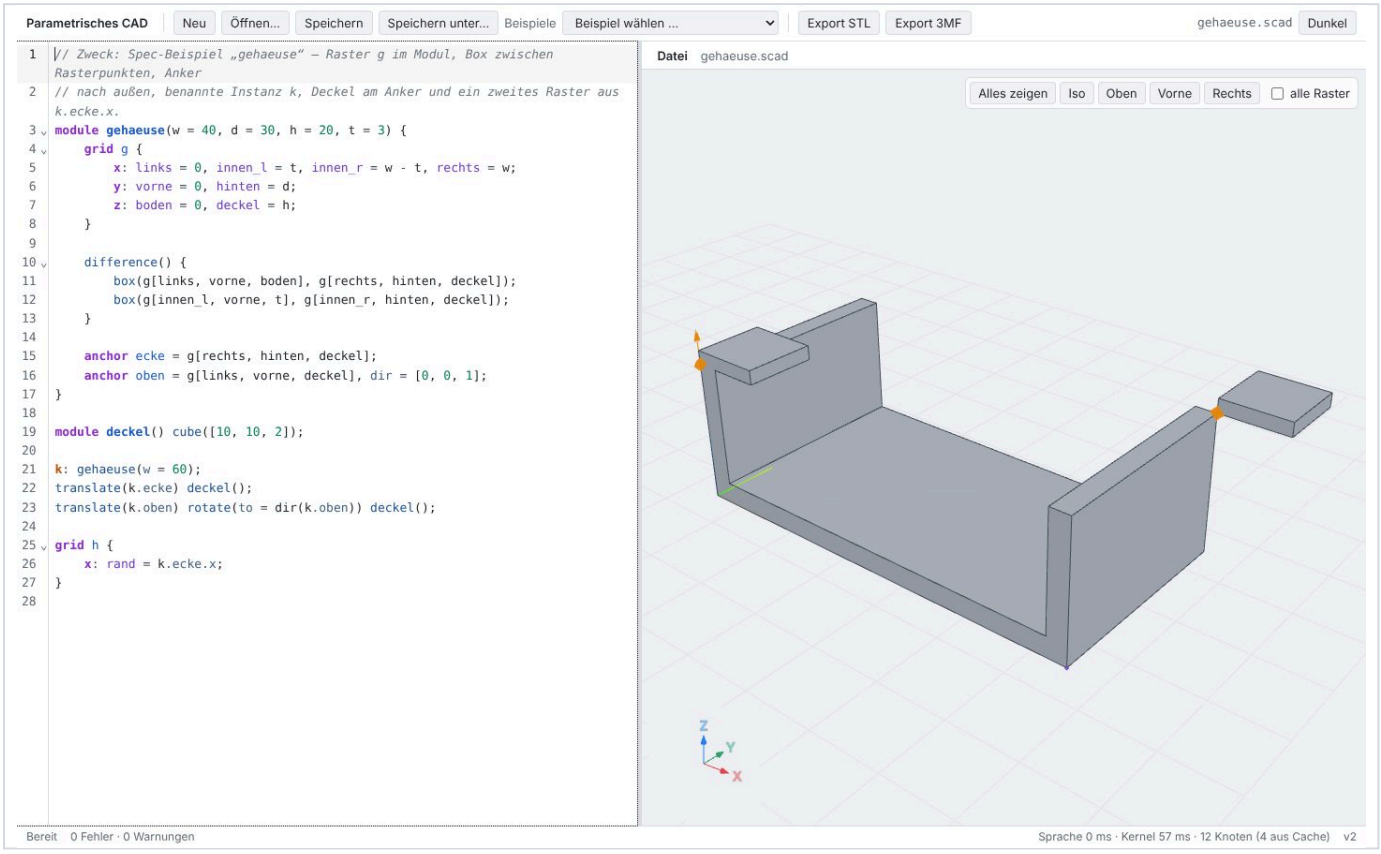
Software rendering without GPU; on a desktop with a GPU the frame rates are considerably higher. The times include the evaluation in the worker (language + kernel) and loading the meshes.

6 Known limitations and deviations

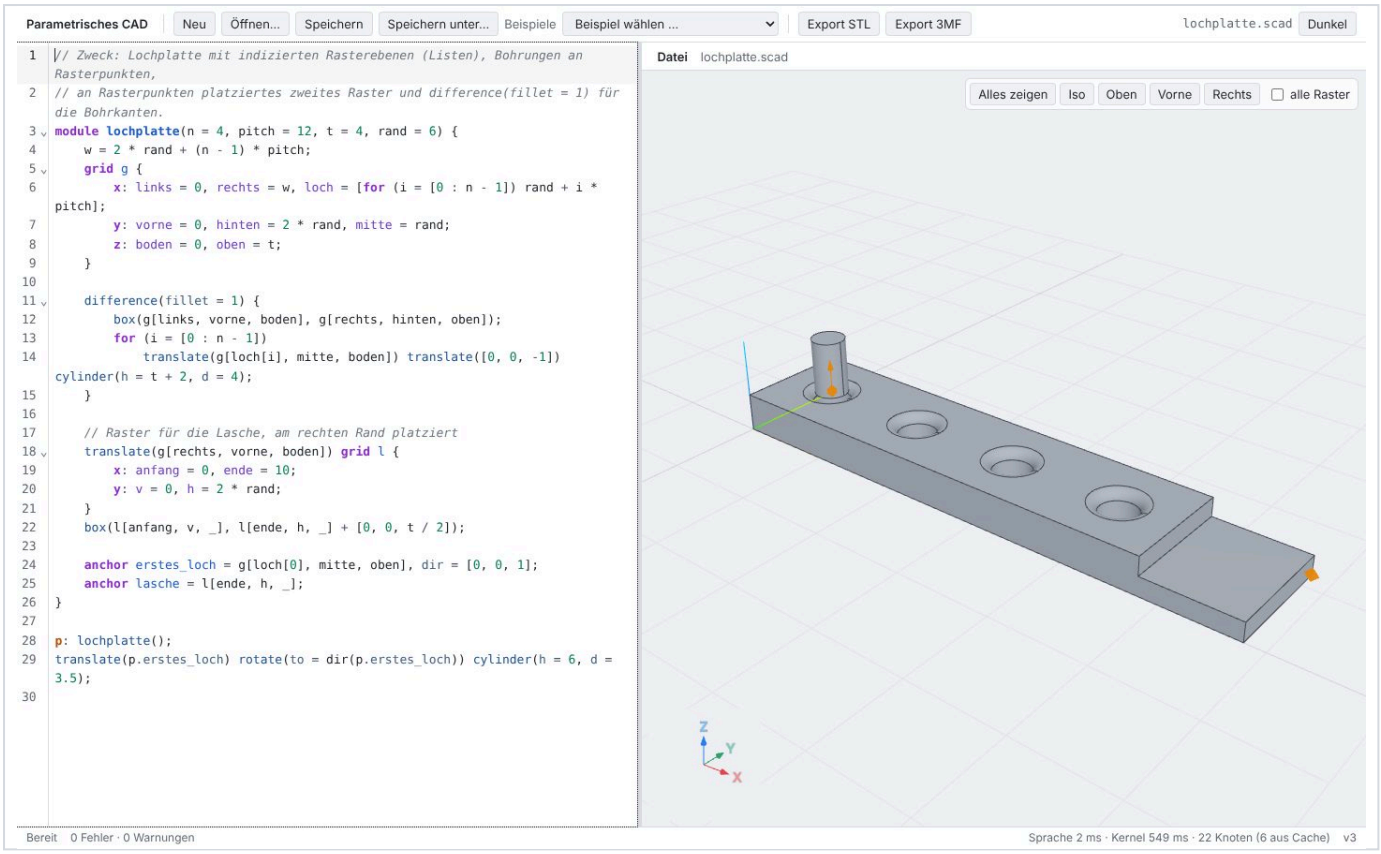
- Handles, dragging, snapping and the parameter bar are not part of phase A; grids and anchors are only displayed.
- Hover over edges is resolved via the adjacent face (edges carry no node tags). Union children get a thin outline on hover, but not with the cursor.
- When the module context changes, the UI resends the source with a new version so that each scene version denotes exactly one evaluation (meshes are requested per version). This costs one additional language evaluation of a few milliseconds.
- In case of ambiguity (same content at several places, e.g. two lids), the location of the hit point within the node bounding boxes is used to disambiguate; if it remains ambiguous, all lines are highlighted.
- Grid expressions in loops (e.g. `g[loch[i], mitte, boden]`) are reported once by the scene; on hover the point of the first iteration is highlighted.
- The editor wraps long lines; hints at the end of a line can therefore move to a second screen line.

- The watchdog test blocks the worker via a test message; a real hanging kernel call was not reproduced.
- The measurements come from a headless Chromium with SwiftShader and are only meaningful relative to each other.

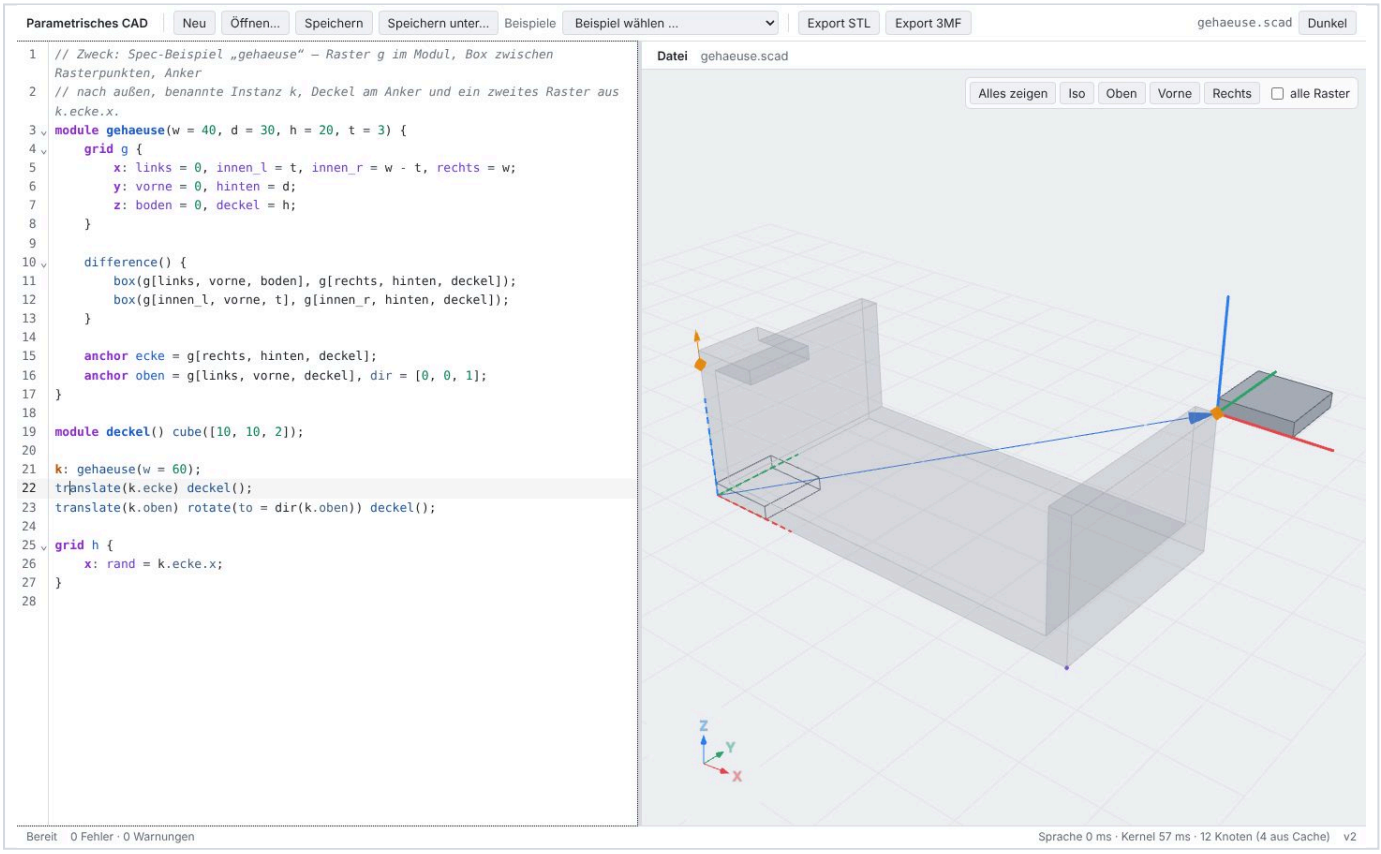
7 Screenshots



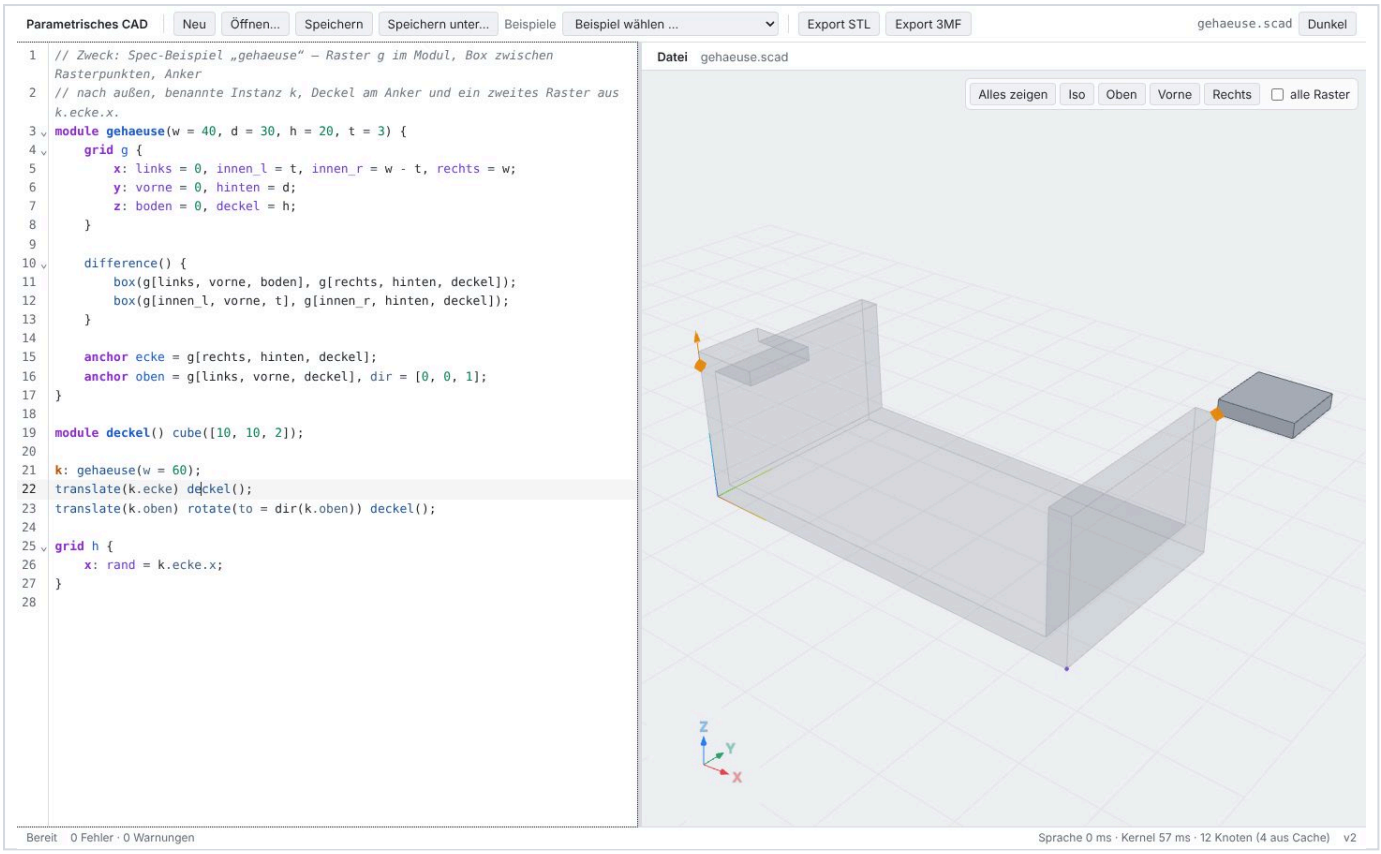
S01 Load. Initial state: spec example gehaeuse.scad in the editor on the left, on the right the scene with ground, axis triad and the anchors of the named instance k (diamonds, with an arrow where dir is set).



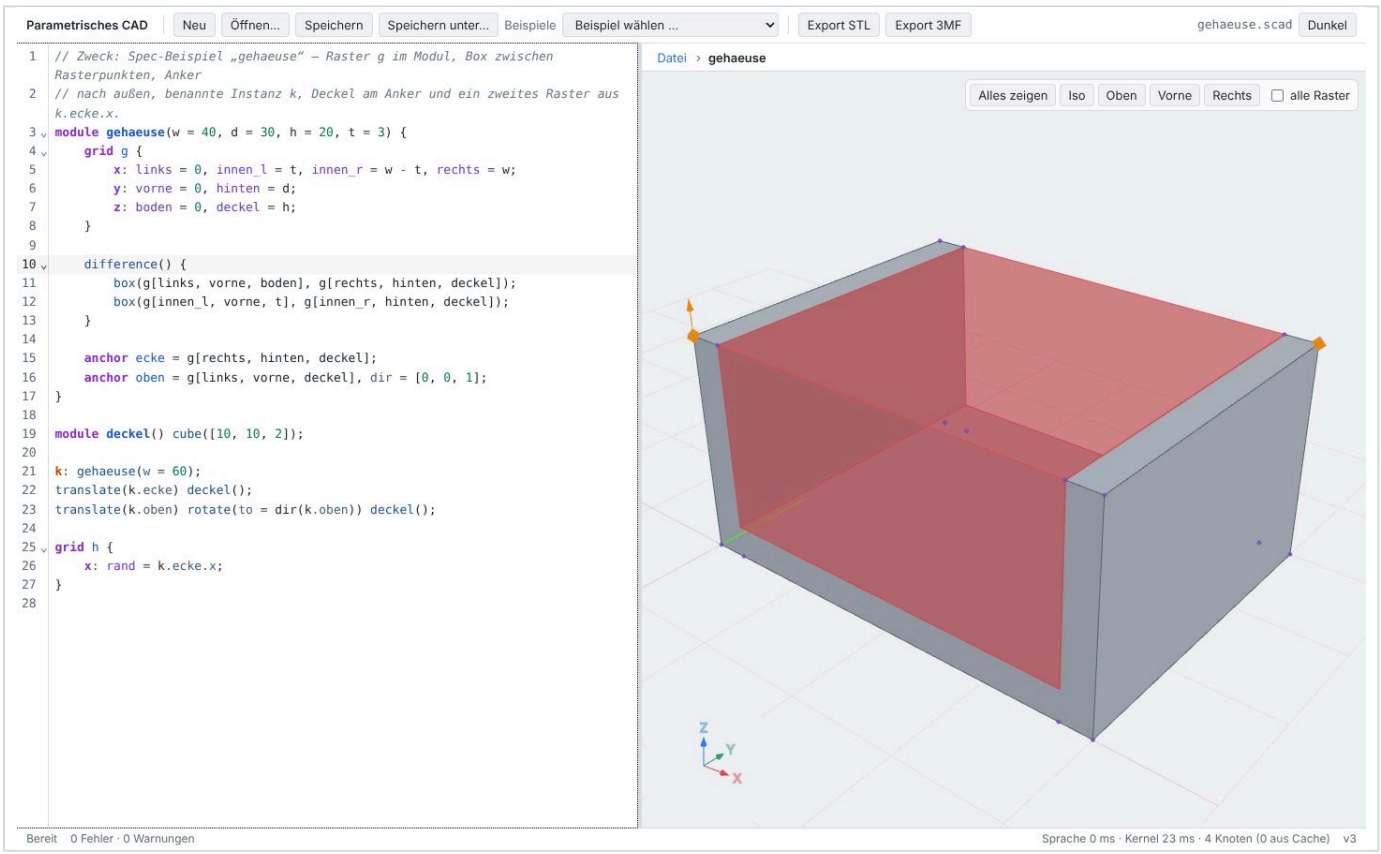
S02 Open example. Examples menu → spec/lochplatte.scad: the editor loads the file, after evaluation the view shows the plate with holes, tab and the anchor p.erstes_loch.



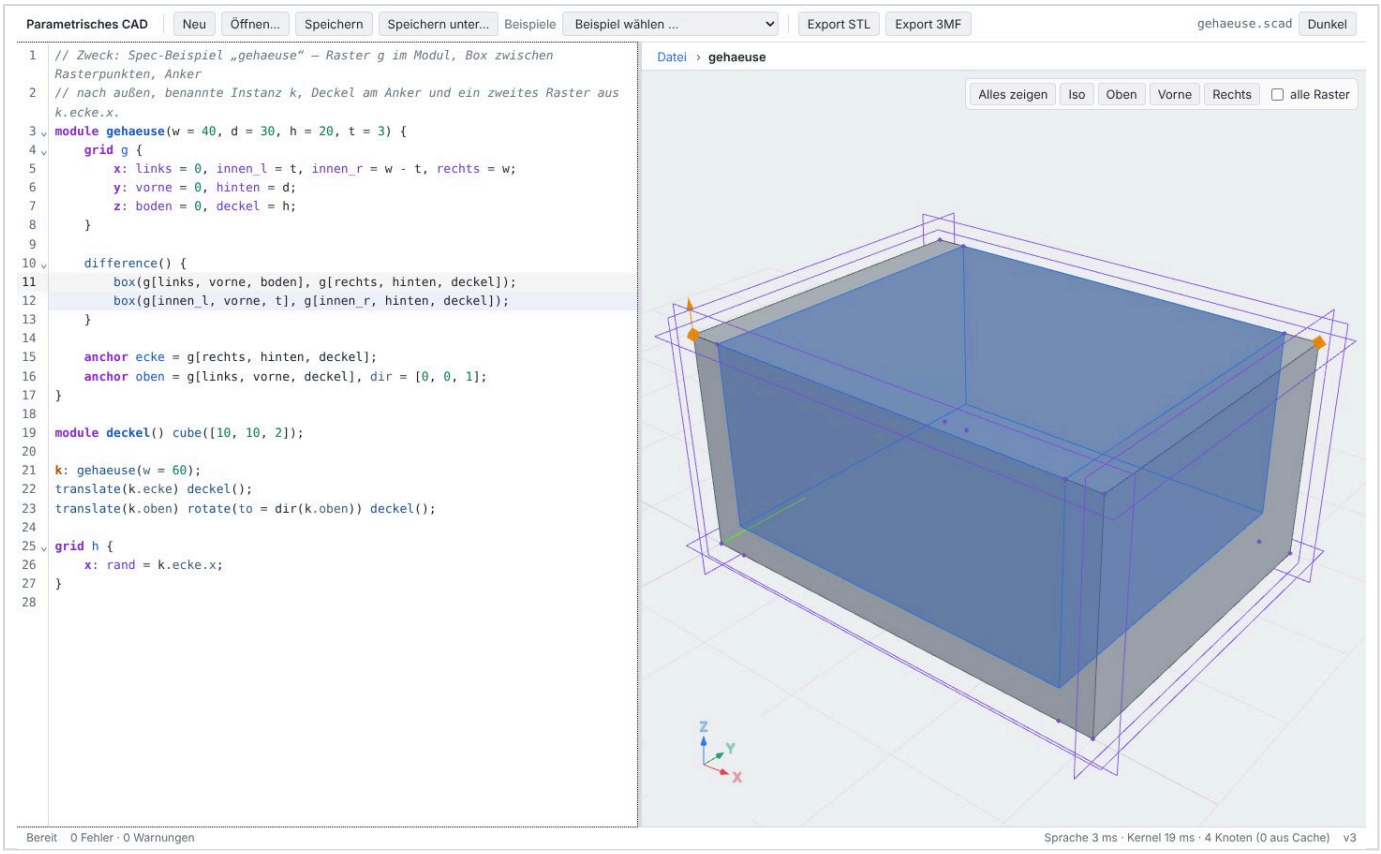
S03 Cursor in line. Cursor on translate(k.ecke) in line 22: subtree full at the target, as outline at the original position; coordinate triad dashed before, solid after, arrow from the old to the new origin; rest of the model dimmed.



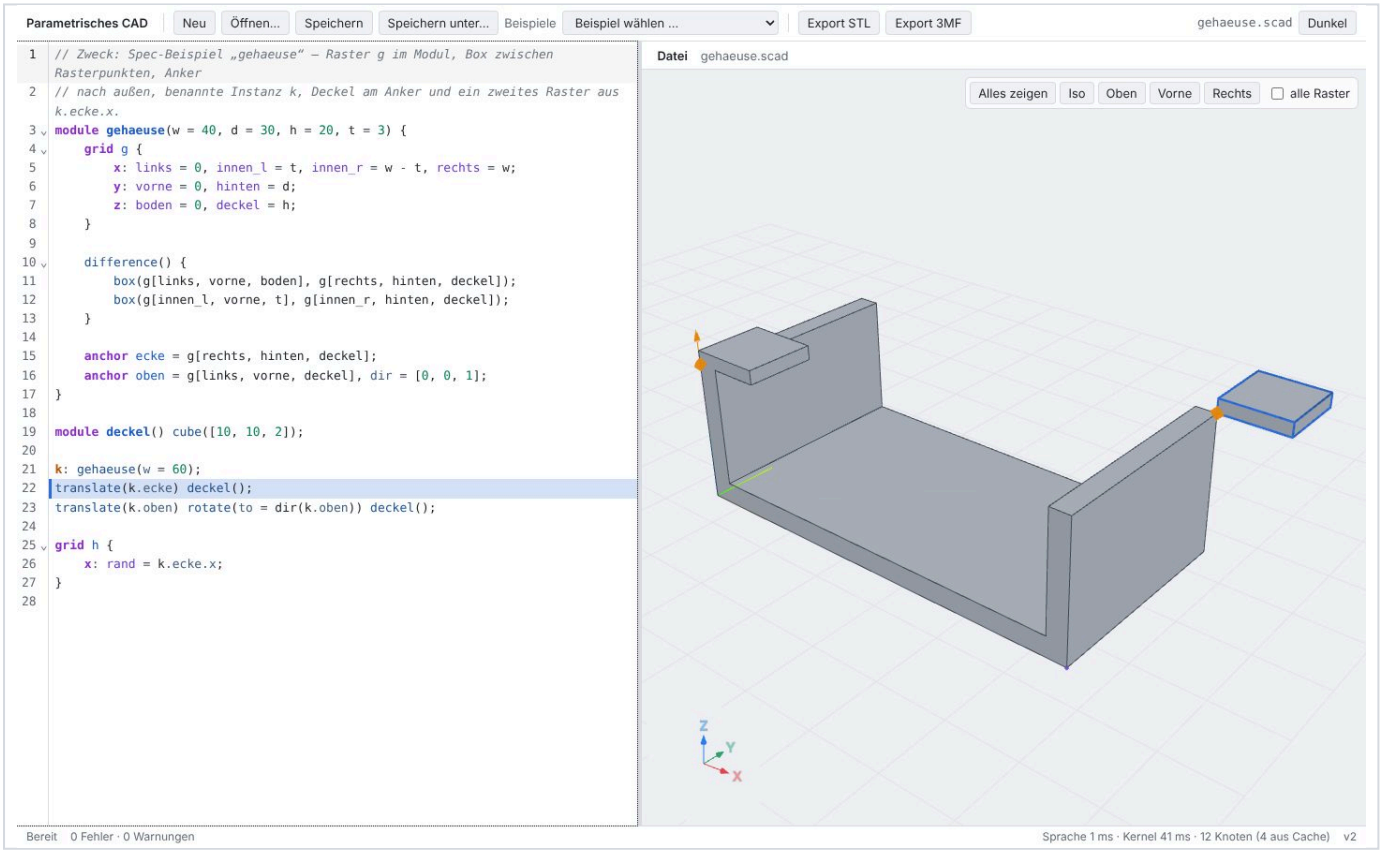
Cursor on deckel() in line 22: module call full, rest dimmed.



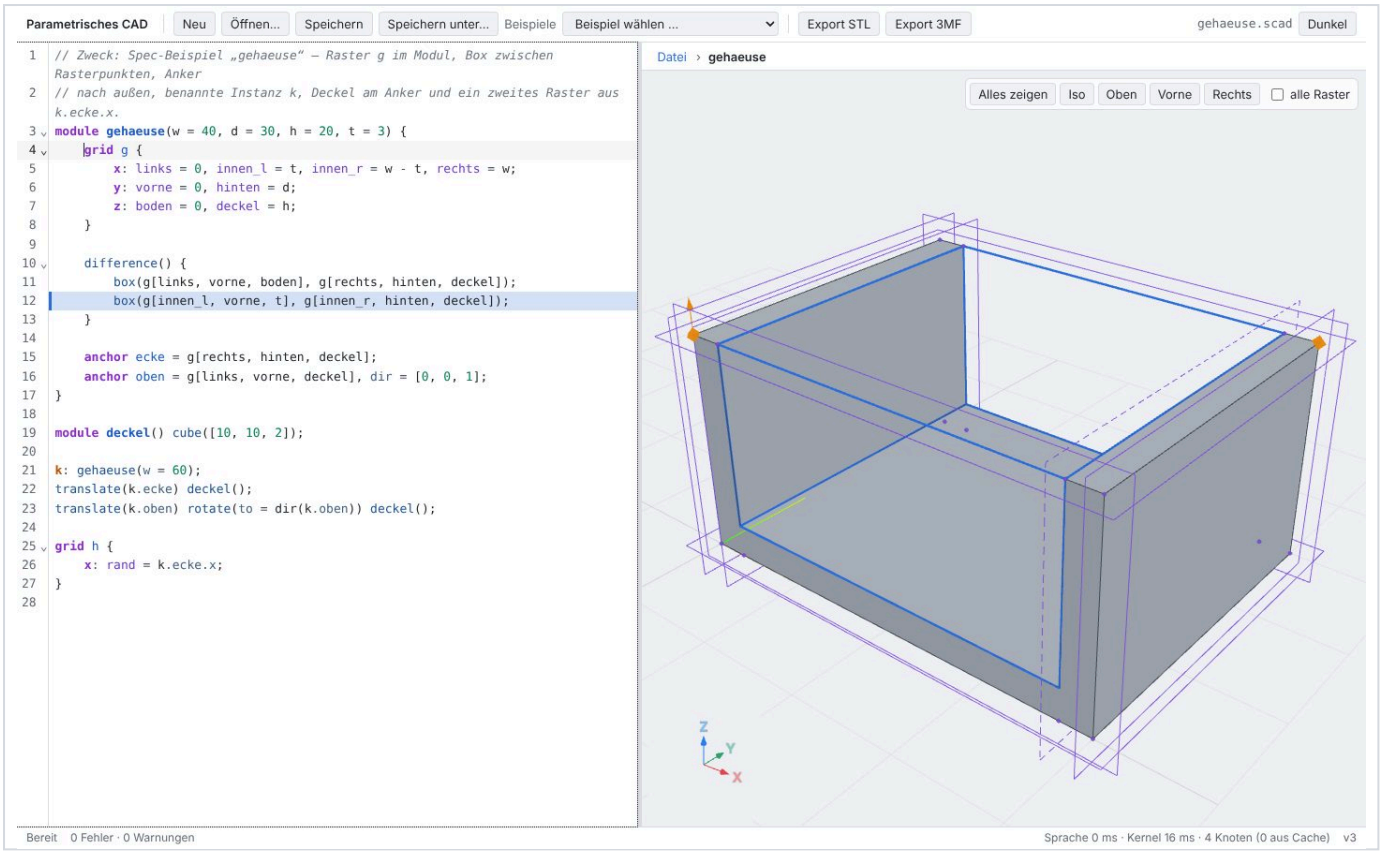
S05 Module context and difference. Cursor in line 10 (difference) inside module gehaeuse: context switches to the module (breadcrumb "File › gehaeuse", own coordinate system, grid g shown); result full, subtracted box from line 12 red and semi-transparent.



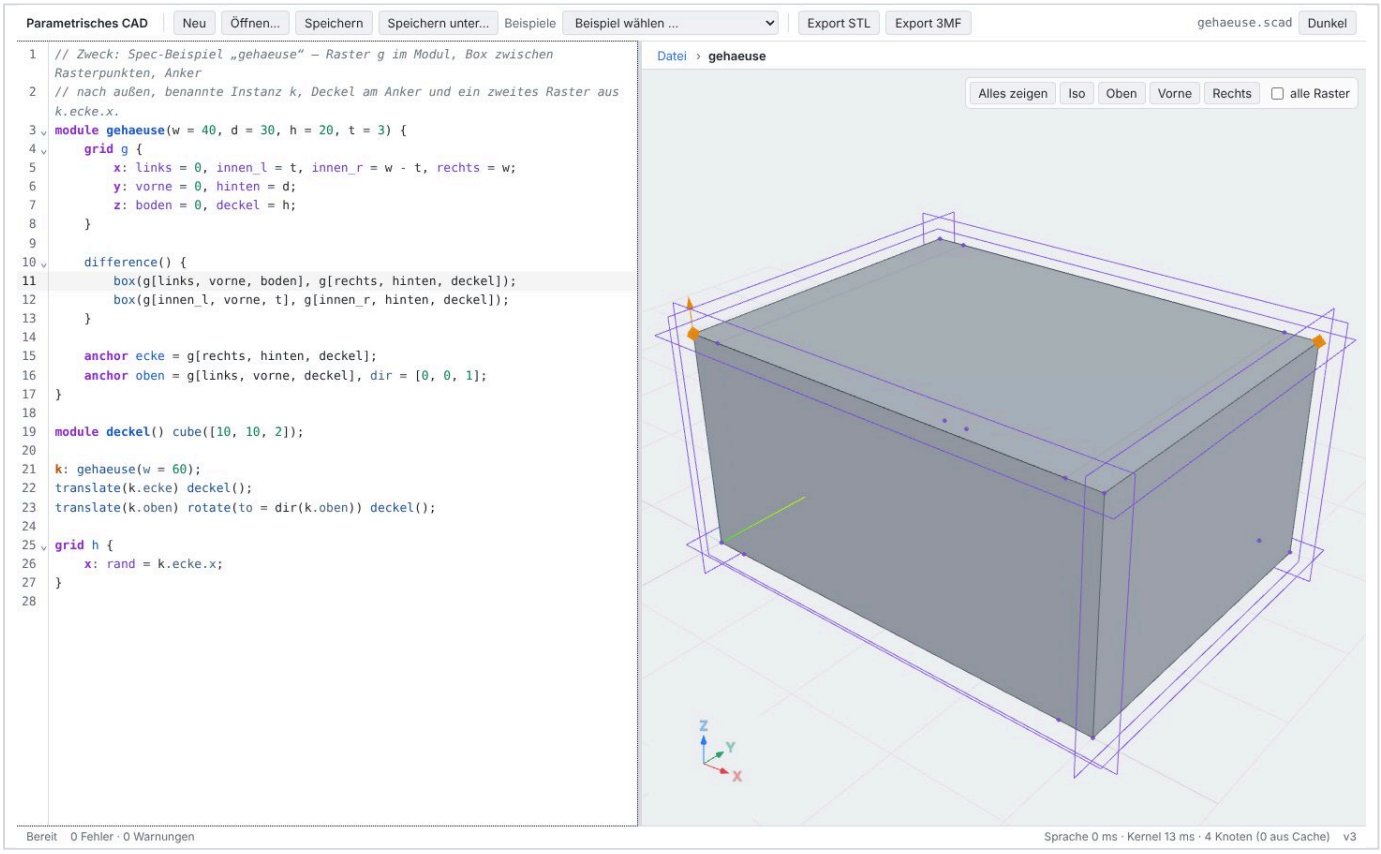
S06 Hover over line. Mouse over line 12 (inner box, subtracted): the body appears semi-transparent in the overall model, visible through the result; cursor is on line 11 (outer box full).



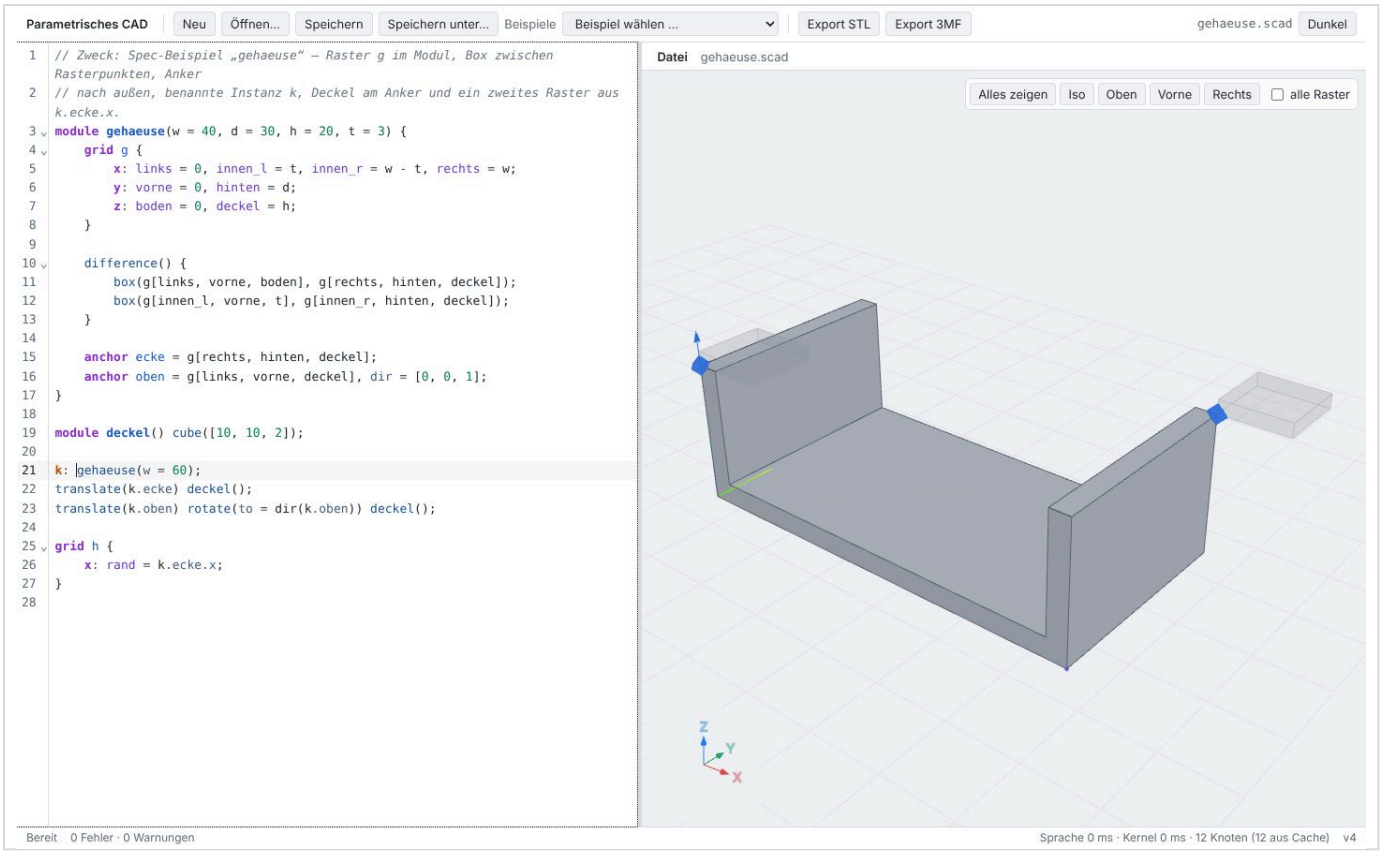
S07 View → code. Mouse over the face of the lid at k.ecke: line 22 in the editor is highlighted (face → tri_tags → tag_nodes → node → line). A click sets the cursor in line 22, a double-click on the enclosure jumps into module gehaeuse.



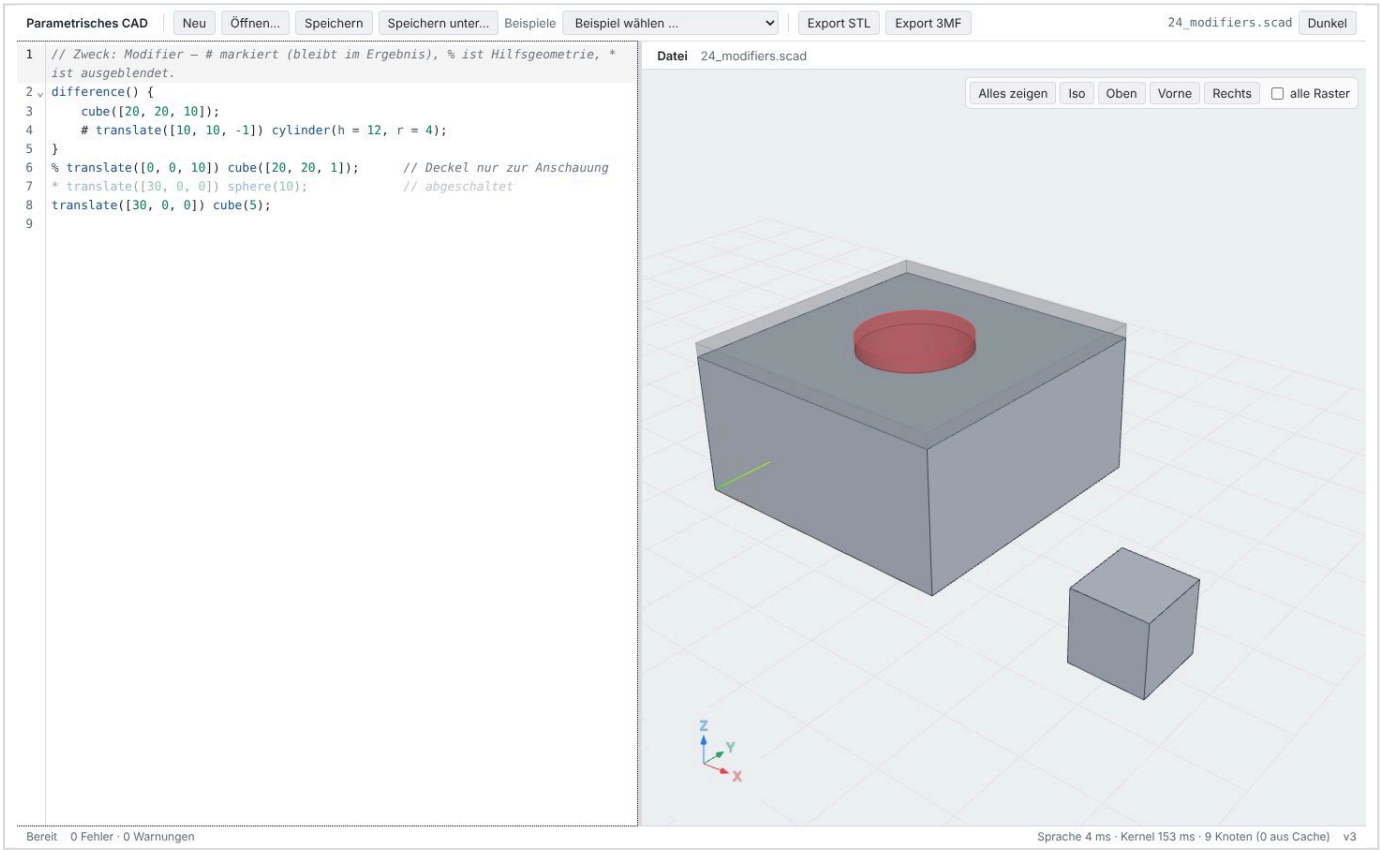
After a double-click on the enclosure: cursor in module gehaeuse, view in module context.



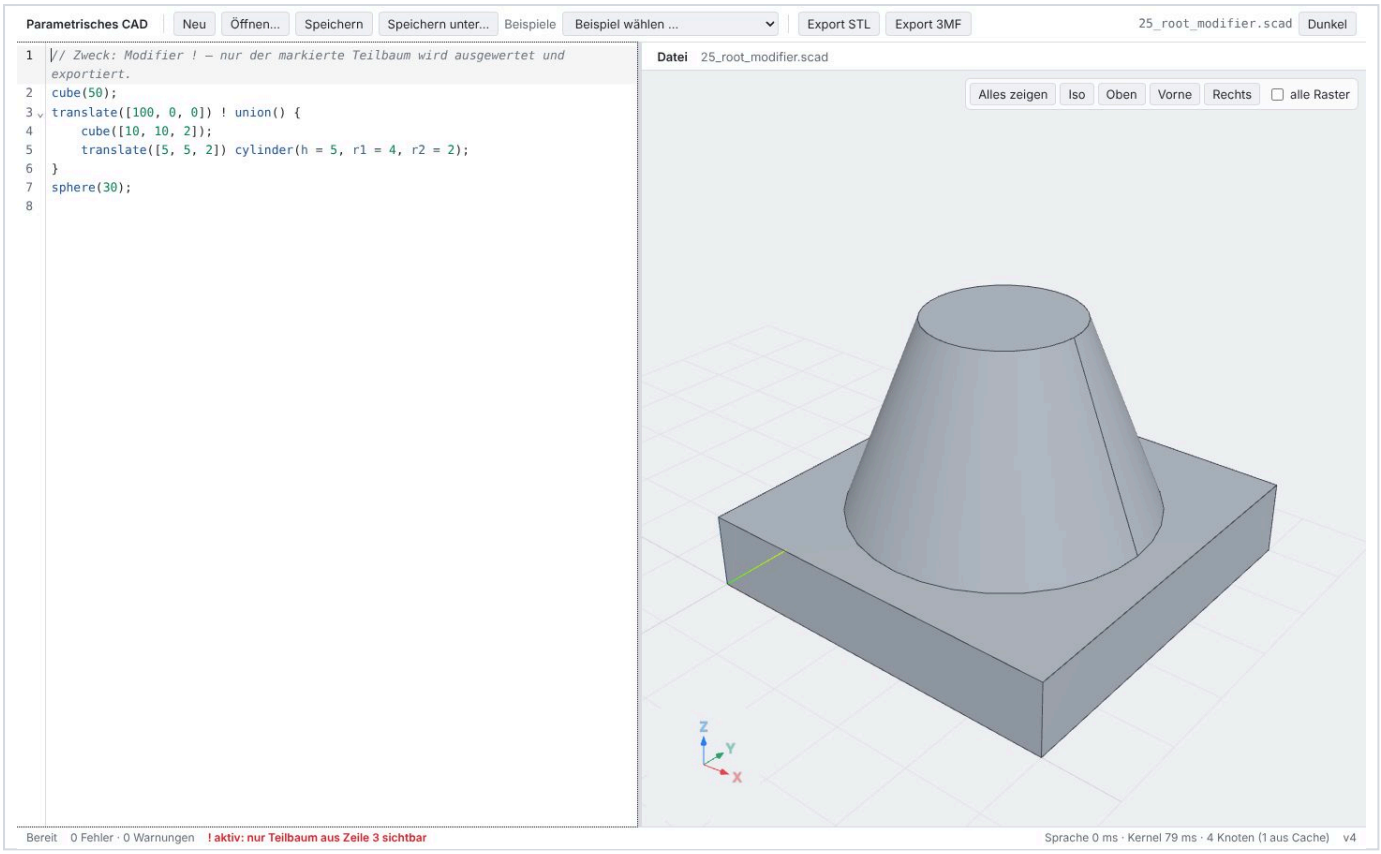
S08 Breadcrumb bar. Module context gehaeuse: only the module in its own coordinate system with grid g (intersection points, planes used by the line). A click on "File" sets the cursor on the call k: gehaeuse(w = 60) (line 21) and shows the whole file again.



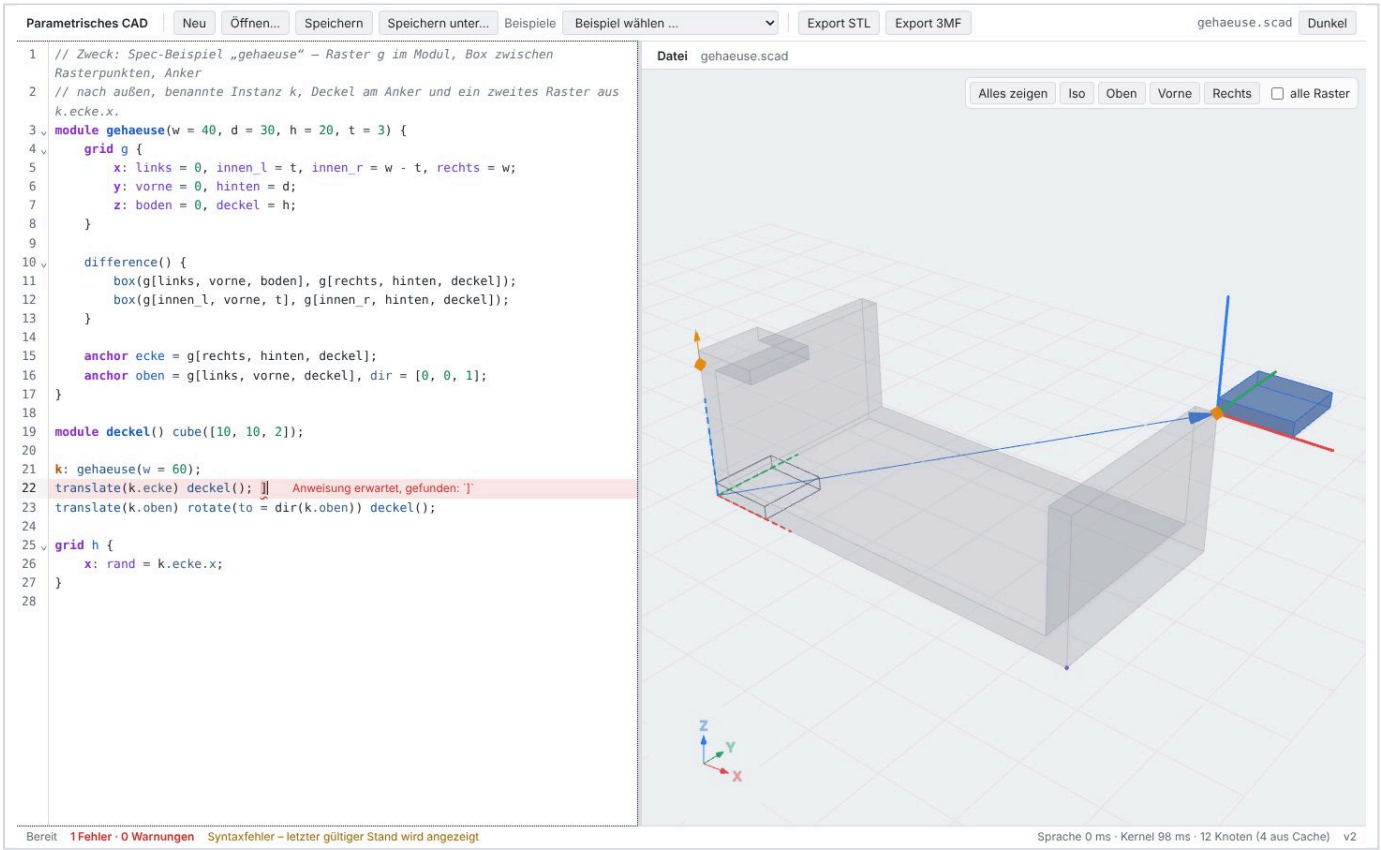
After a click on "File" in the breadcrumb bar: cursor on k: gehaeuse(w = 60), whole file.



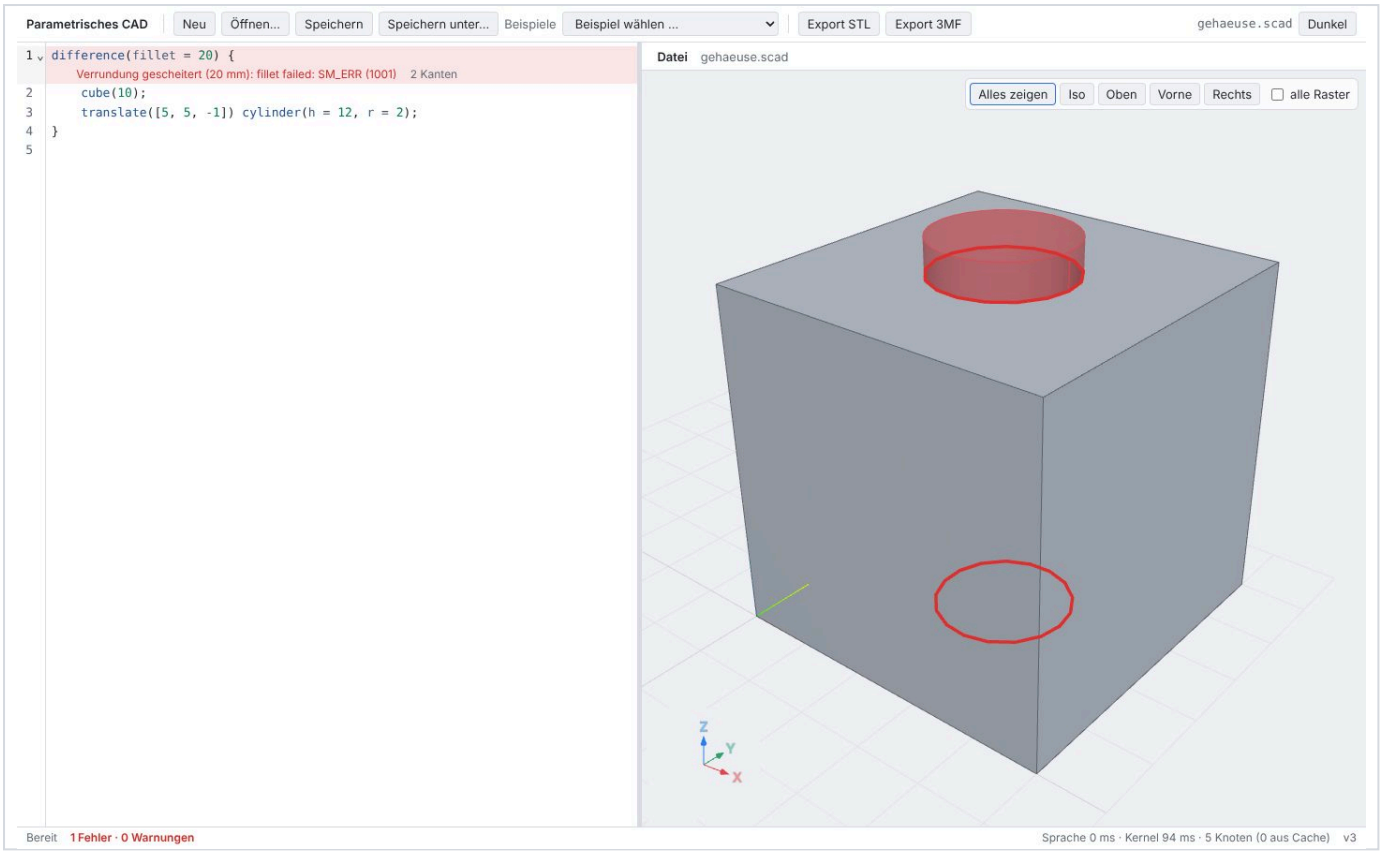
S09 Modifiers. 24_modifiers.scad: # permanently red and semi-transparent (also as a subtracted body), % gray and semi-transparent as helper geometry, * invisible and line 7 grayed out. With 25_root_modifier.scad the status bar reports the active ! subtree.



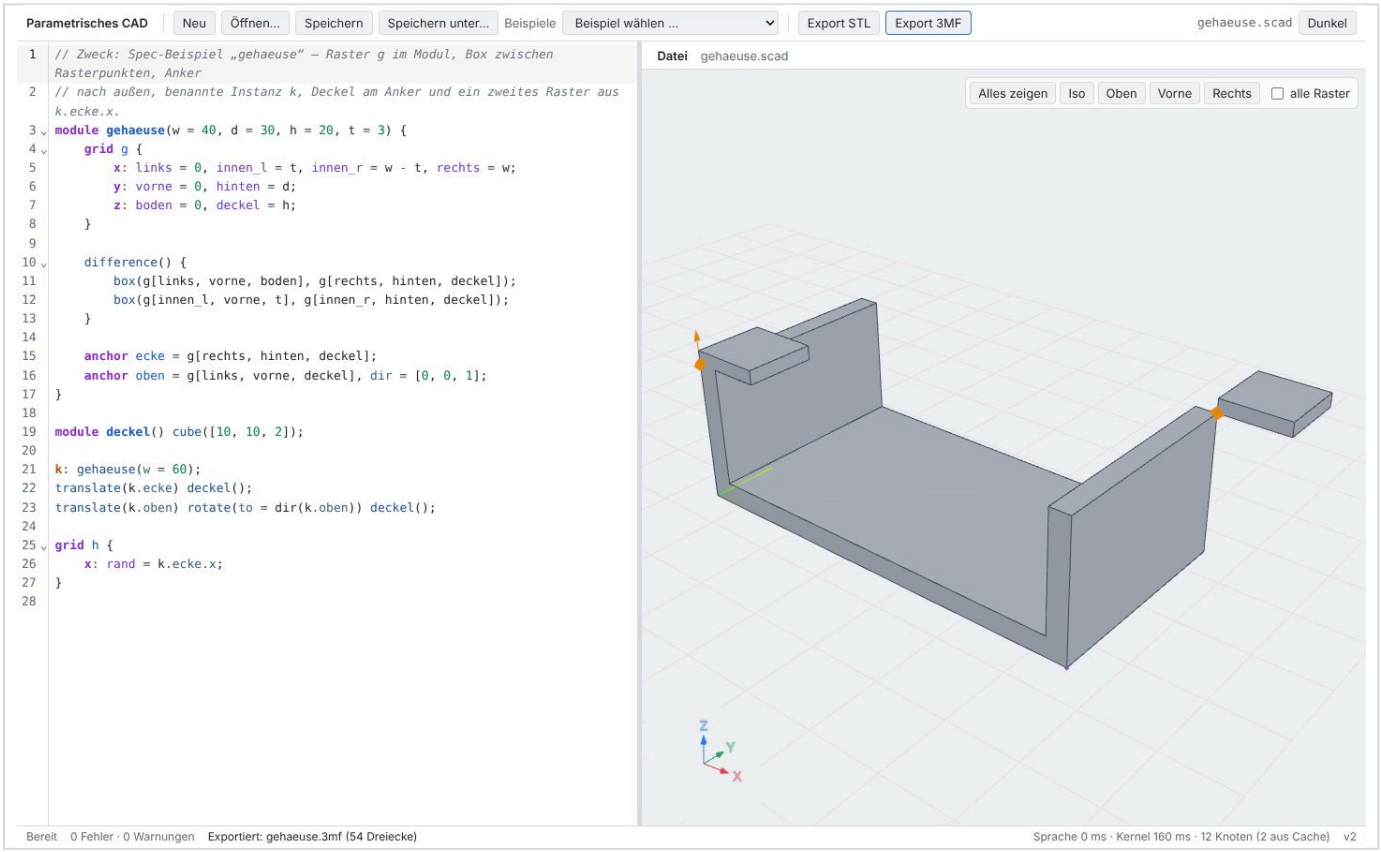
25_root_modifier.scad: only the ! subtree is shown, hint in the status bar.



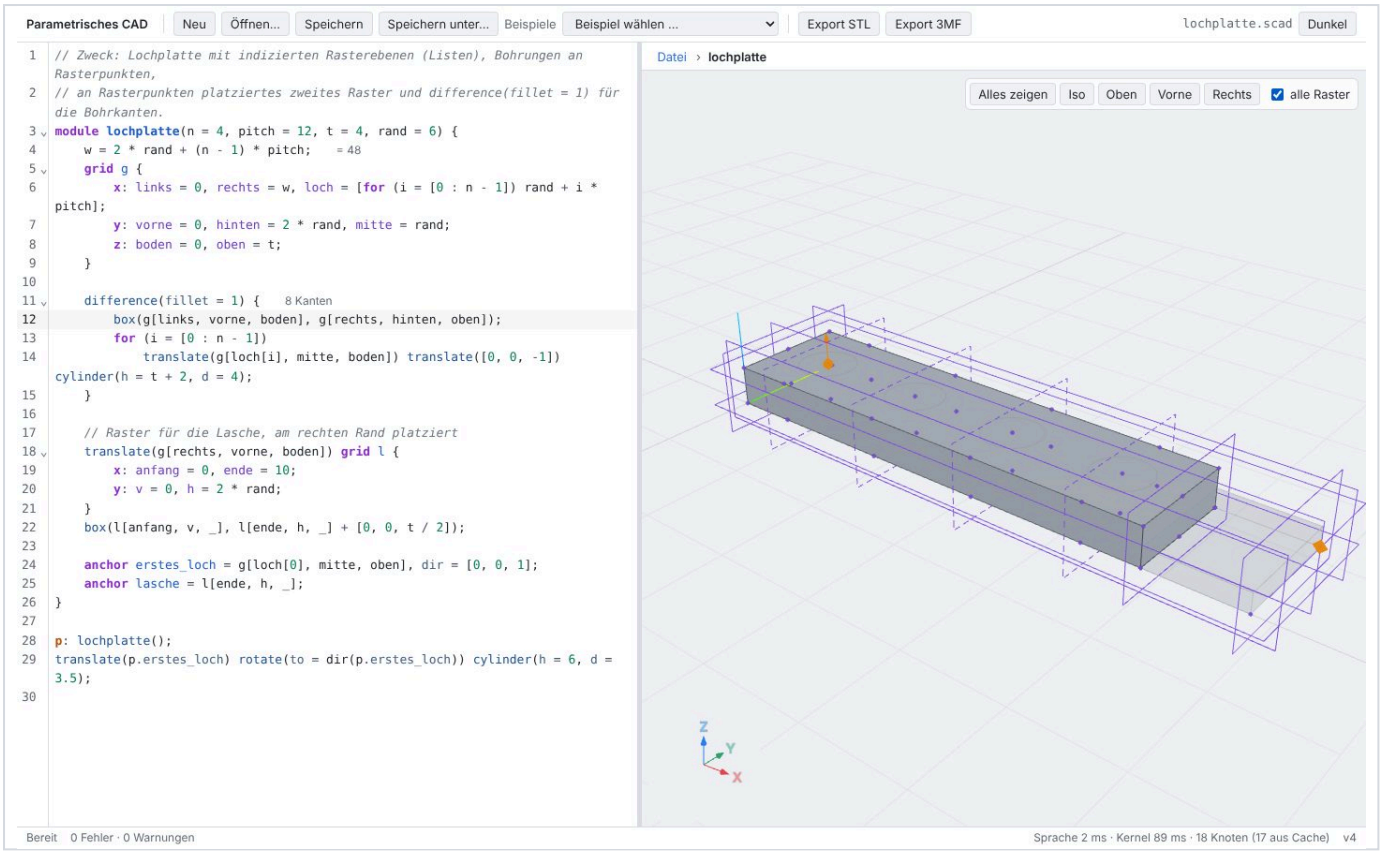
S10 Syntax error. Input "]" at the end of line 22: position underlined red, message at the end of the line, status bar "1 error"; the view keeps showing the last valid state. After deleting it, the model is re-evaluated.



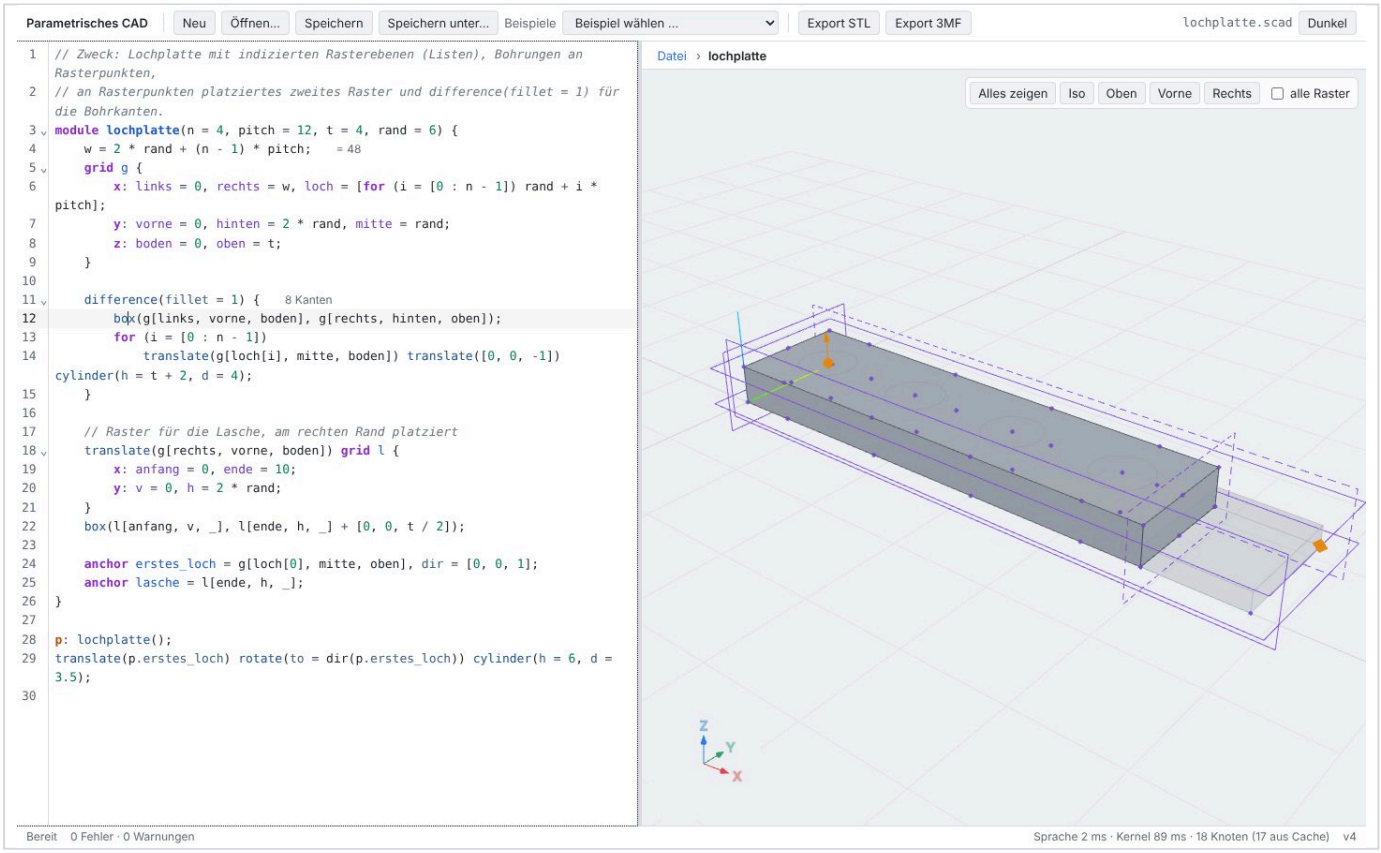
S11 Failed operation. difference(fillet = 20) on a 10 mm cube: fillet failed, line 1 red with a message at the end of the line; the view shows the body without fillet.



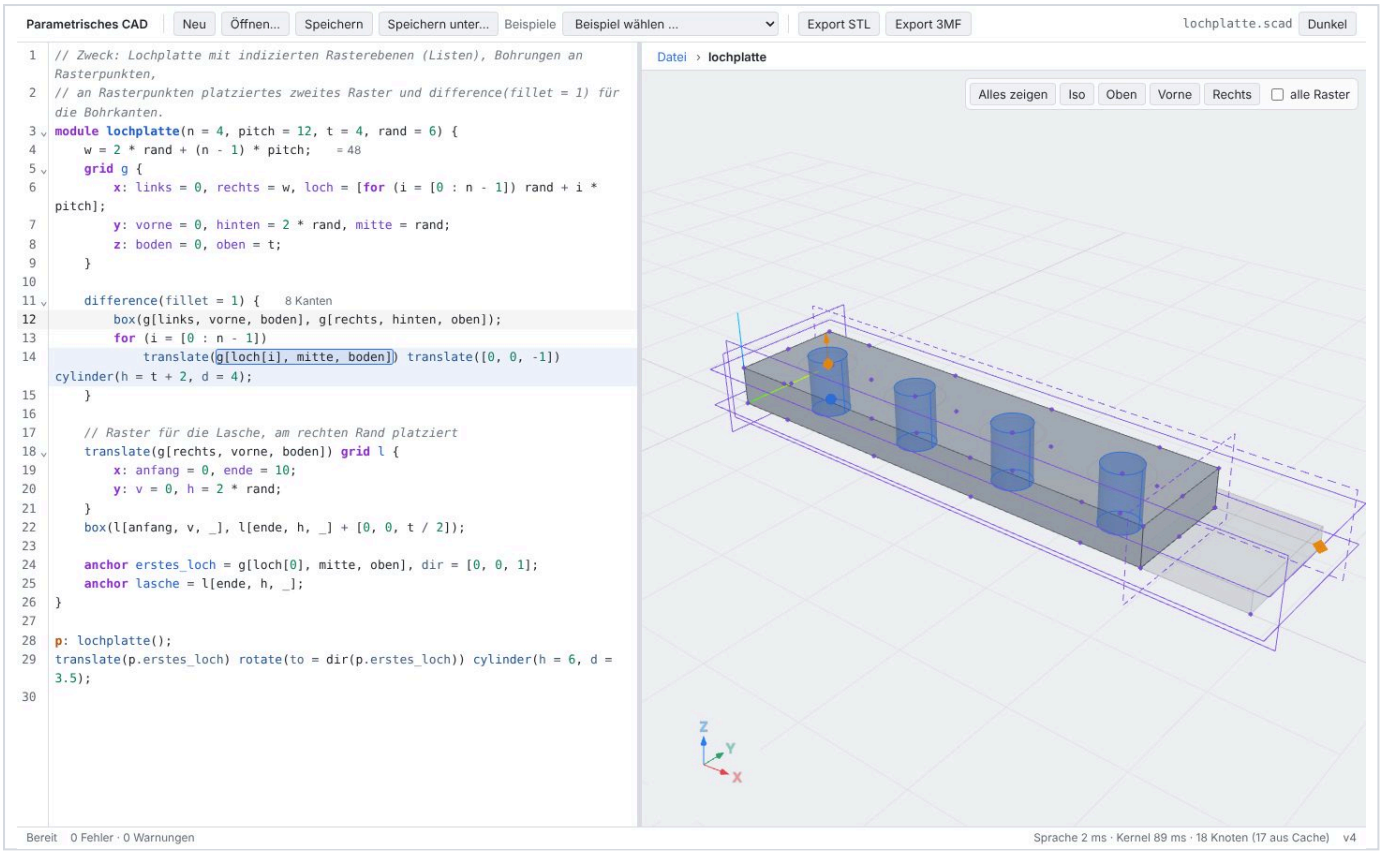
S12 Export. Export STL: gehaeuse.stl, 54 triangles, 2784 bytes (binary, 84 + 50·n). Export 3MF: gehaeuse.3mf (ZIP, 4486 bytes). The status bar reports the export.



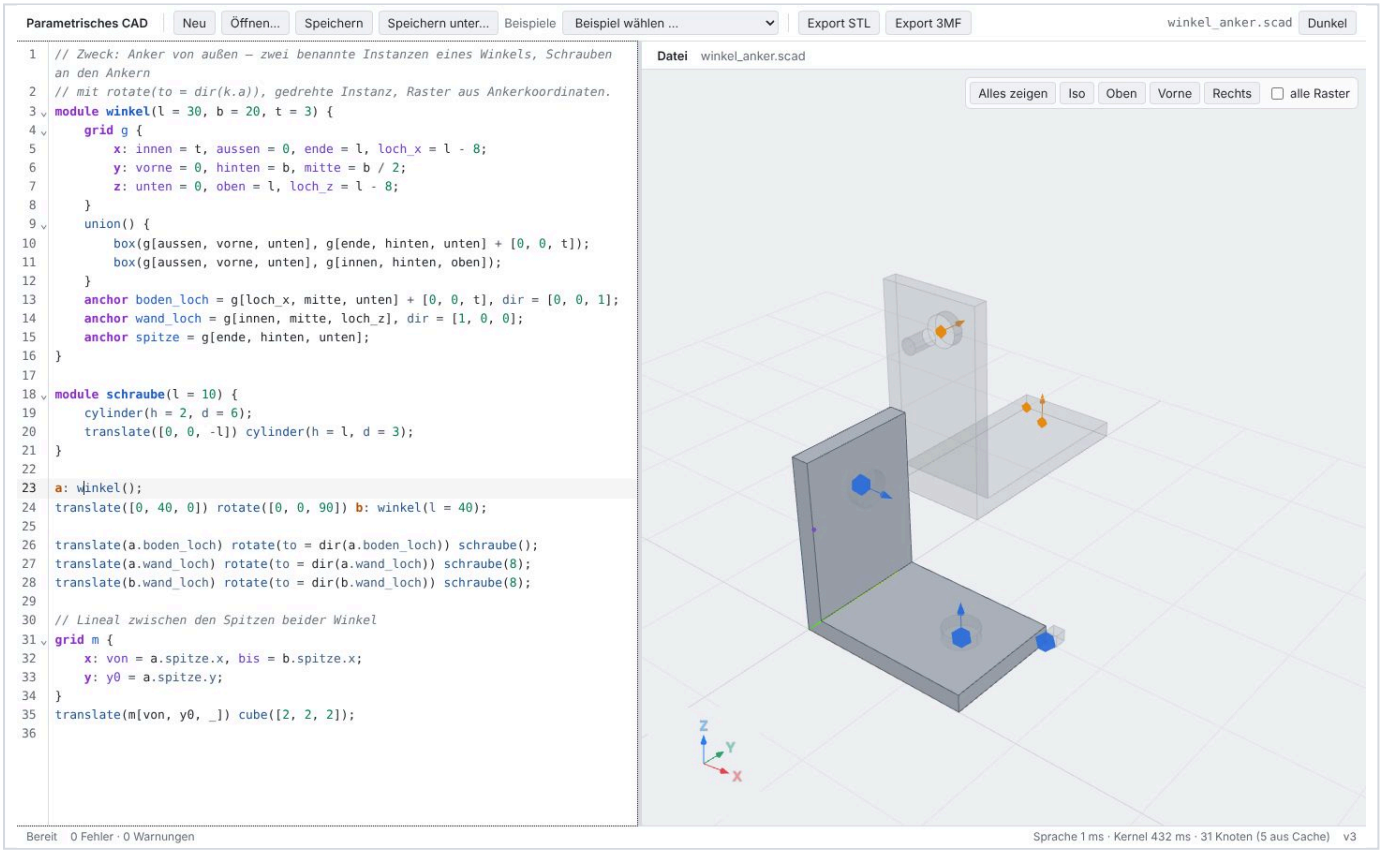
S13 Grid. Module lochplatte, toggle "all grids": planes as thin lines at the edge of the part's bounding box (literal solid, expression dashed), intersection points as dots. Without the toggle only the 6 planes of line 12; hovering over g[loch[i], mitte, boden] highlights the point.



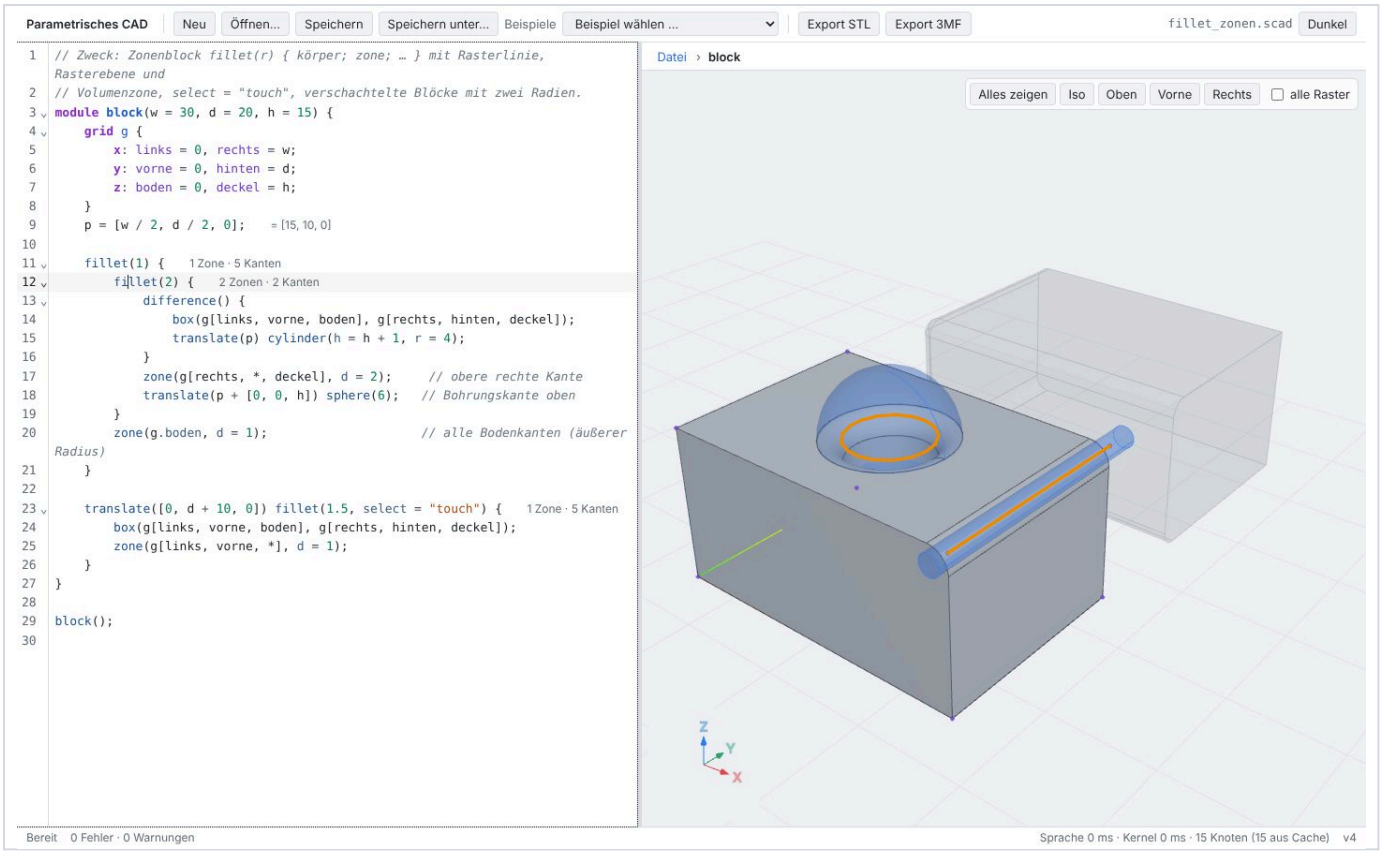
Grid with only the planes of line 12 (box between grid points).



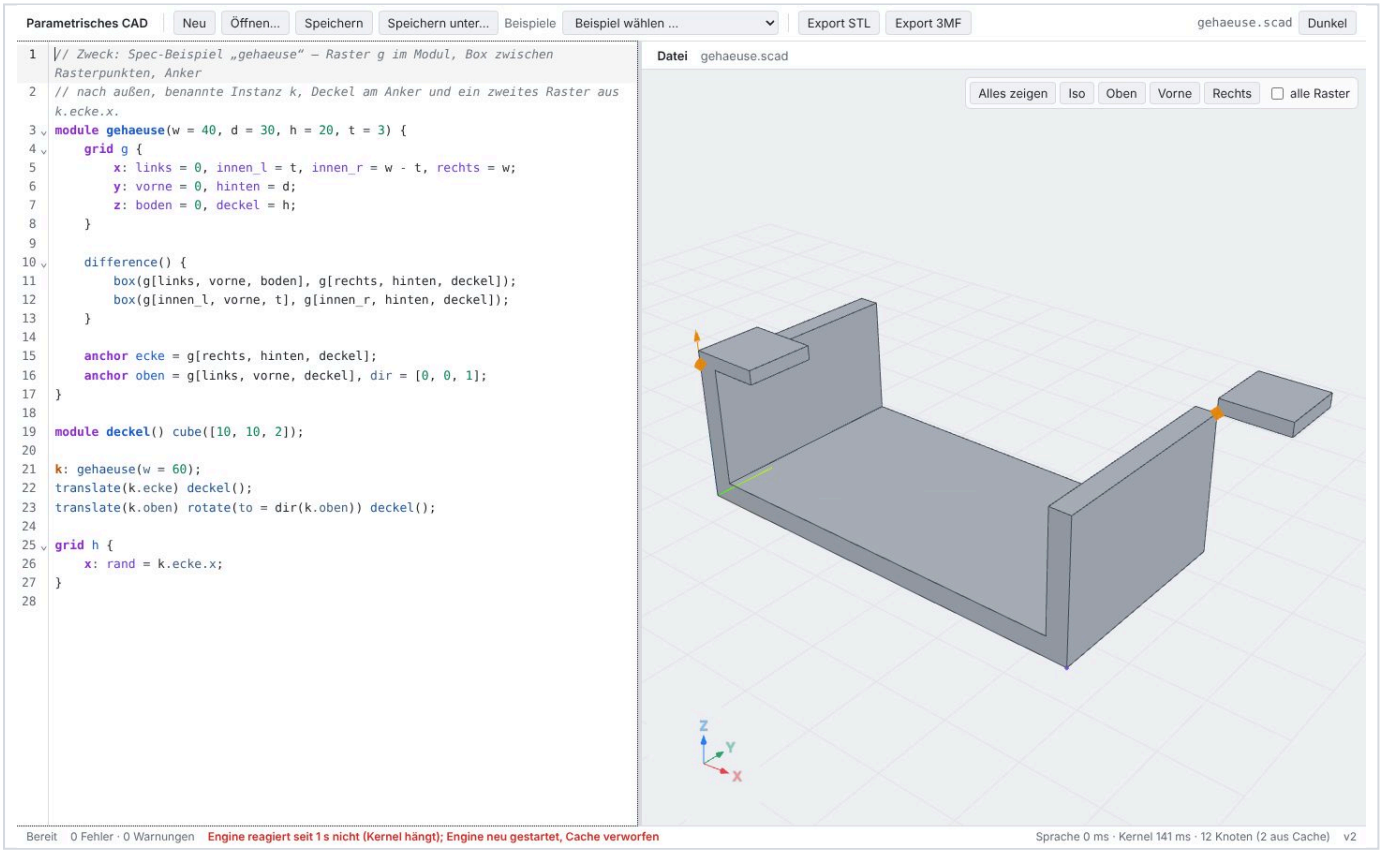
Hover over g[loch[i], mitte, boden]: expression highlighted, grid point highlighted (in loops, the point of the first iteration).



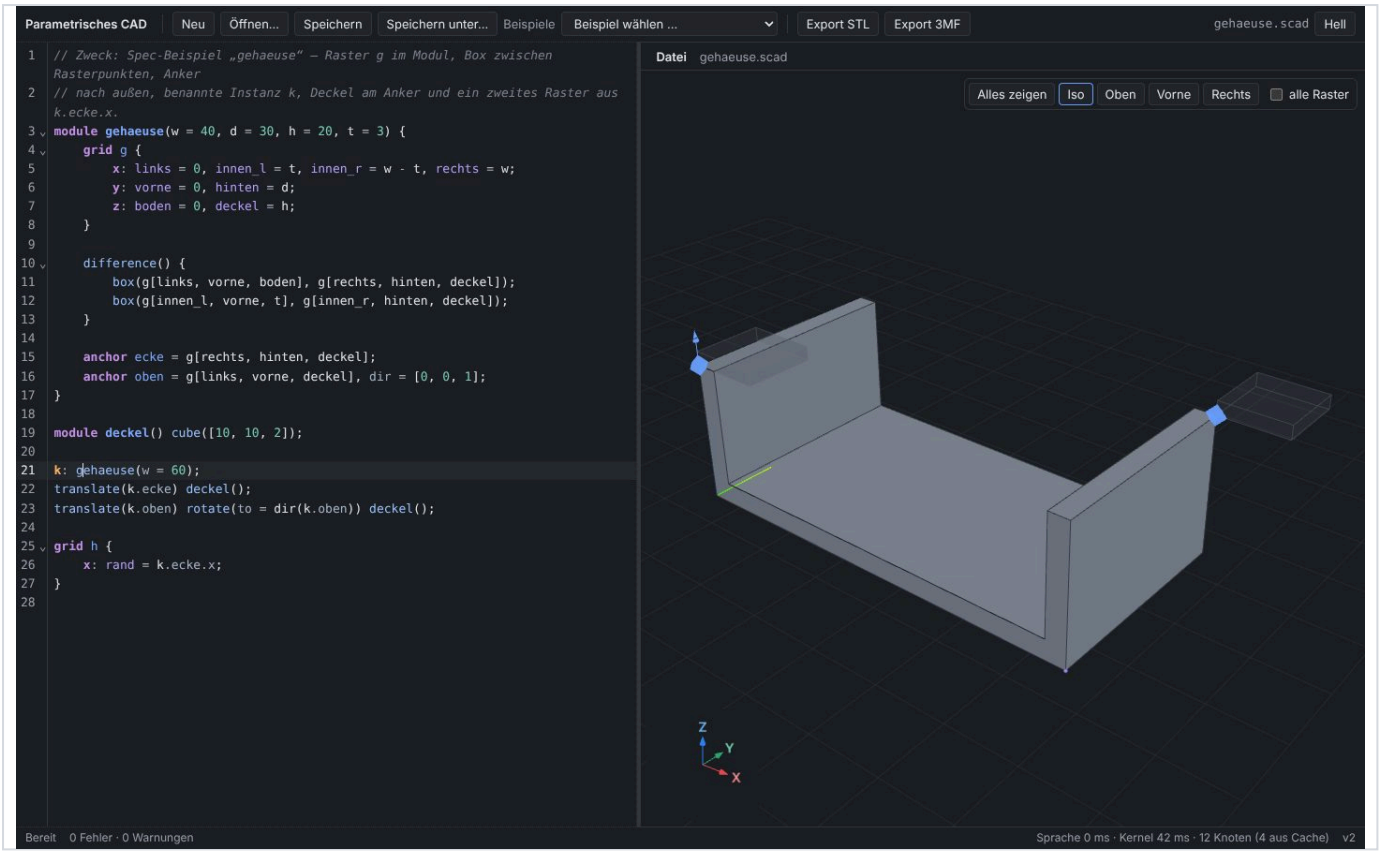
S14 Anchors. winkel_anker.scad: anchors of the named instances a and b as diamonds, with an arrow where dir is set. Hovering over a.boden_loch highlights this anchor; the cursor on a: winkel() highlights the three anchors of the instance.



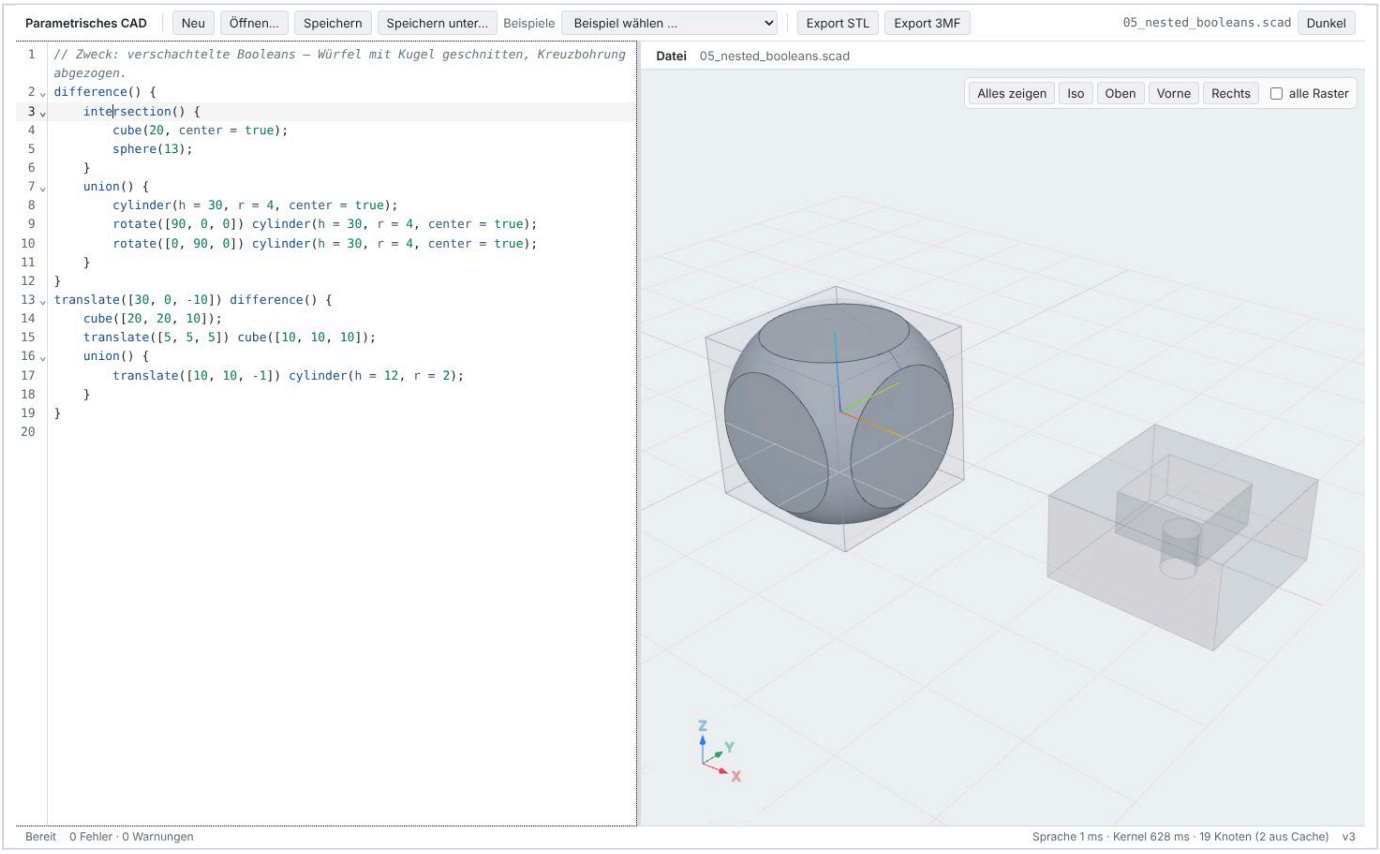
S15 Fillet with zones. fillet_zonen.scad, cursor on fillet(2) in line 12: zones (grid line and sphere) blue and semi-transparent, selected edges orange; inline hint "2 zones · 2 edges" at the end of the line.



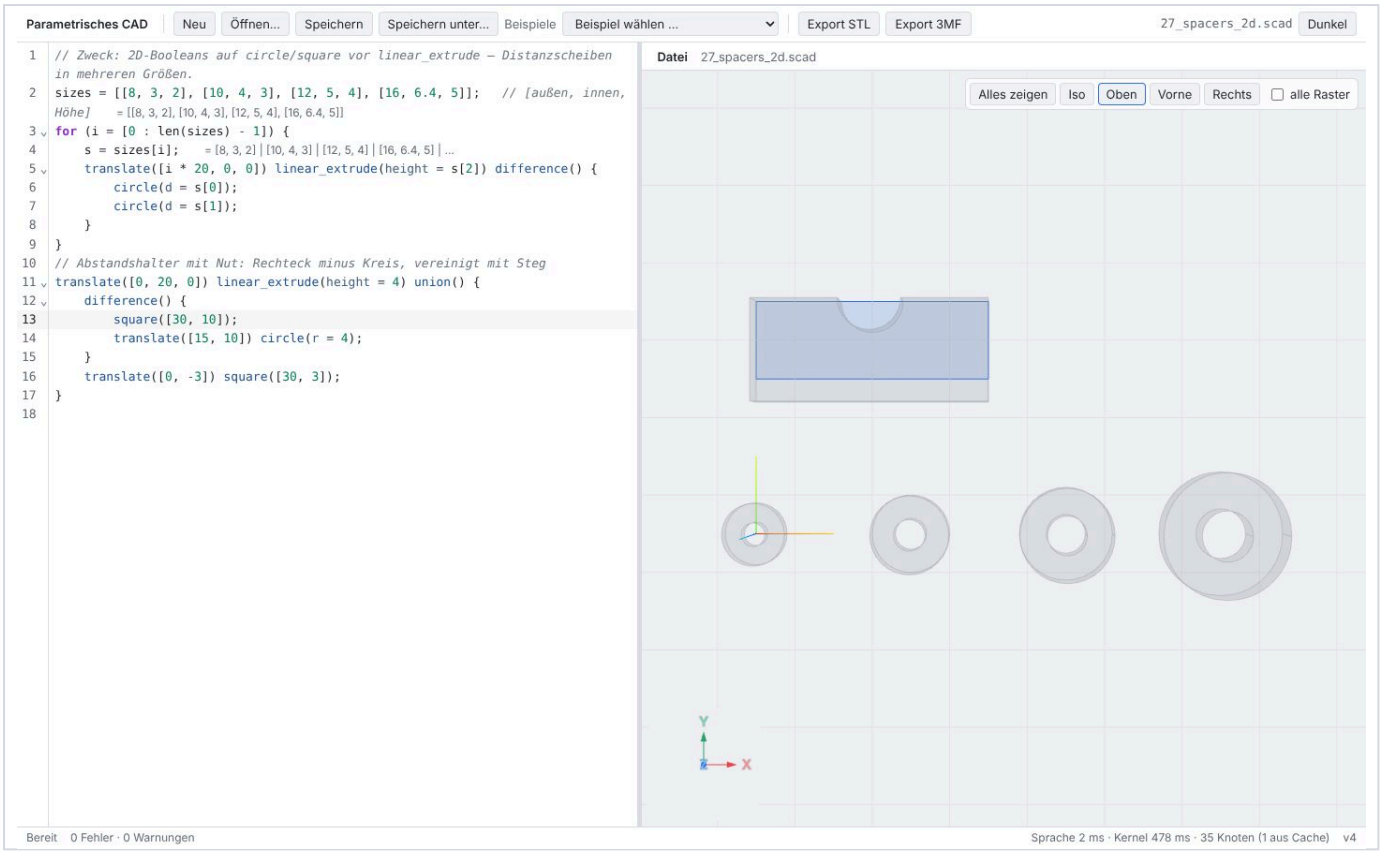
S17 Watchdog. Engine worker blocked for 5 s (test helper, watchdog 0.8 s): the worker is terminated and restarted, the status bar reports the error in red; the last scene stays visible, the next change is evaluated normally.



S19 Dark theme. Dark theme, iso view, cursor on k: gehaeuse(w = 60): instance full, anchors of the instance highlighted, rest dimmed.



S20 intersection and 2D. 05_nested_booleans.scad, cursor on intersection(): cube and sphere semi-transparent, result full. 27_spacers_2d.scad, cursor on square([30, 10]): 2D shape as outline with a light fill in its plane (top view).



27_spacers_2d.scad, top view: cursor on square([30, 10]) – 2D shape as outline with a light fill, rest dimmed.